Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	The project follows consistent Python and HTML/CSS indentation standards.
2	Proper File Structure	Files and folders are logically organized	Yes	Source code is organized into templates, static, models, utils, and dataset directories.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names are used for variables, datasets, models, and input fields.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly describe their purpose, such as preprocessing, training, prediction, and validation.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections of the code include comments for better understanding and maintenance.

6	Modular Design	Code is split into reusable functions/modules	Yes	The project is divided into reusable modules for preprocessing, model loading, training, feature engineering, and prediction.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated logic is minimized by using reusable functions and helper modules.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation, model loading checks, and exception handling are implemented to improve reliability.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	preprocessing.py	Python	Data cleaning, preprocessing, feature transformation	High
2	model_loader.py	Python	Loading trained model, scaler, encoder, and feature columns	High
3	model_trainer.py	Python	Training, evaluation, and model comparison	High
4	Flask Templates (base.html)	HTML, CSS, Jinja2	Shared layout for Home, Prediction, and Result pages	High
5	JavaScript Validation (predict.js)	JavaScript	Client-side form validation for prediction inputs	Medium

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		modular architecture, and maintainable design suitable for academic and real-world applications.

Overall Code Quality Assessment

Aspect	**Rating (1–5)**	**Comments**
Code Layout & Structure	5	Well-organized folder structure with clear separation of frontend, backend, and machine learning modules.
Readability	5	Code follows consistent naming conventions, indentation, and formatting, making it easy to understand.
Reusability	5	Modular components and reusable helper functions minimize duplication and simplify maintenance.
Documentation / Comments	4	Important logic is documented with comments, and the project includes README and supporting documentation.
Overall Score	**4.8 / 5**	**The project demonstrates good coding practices,**